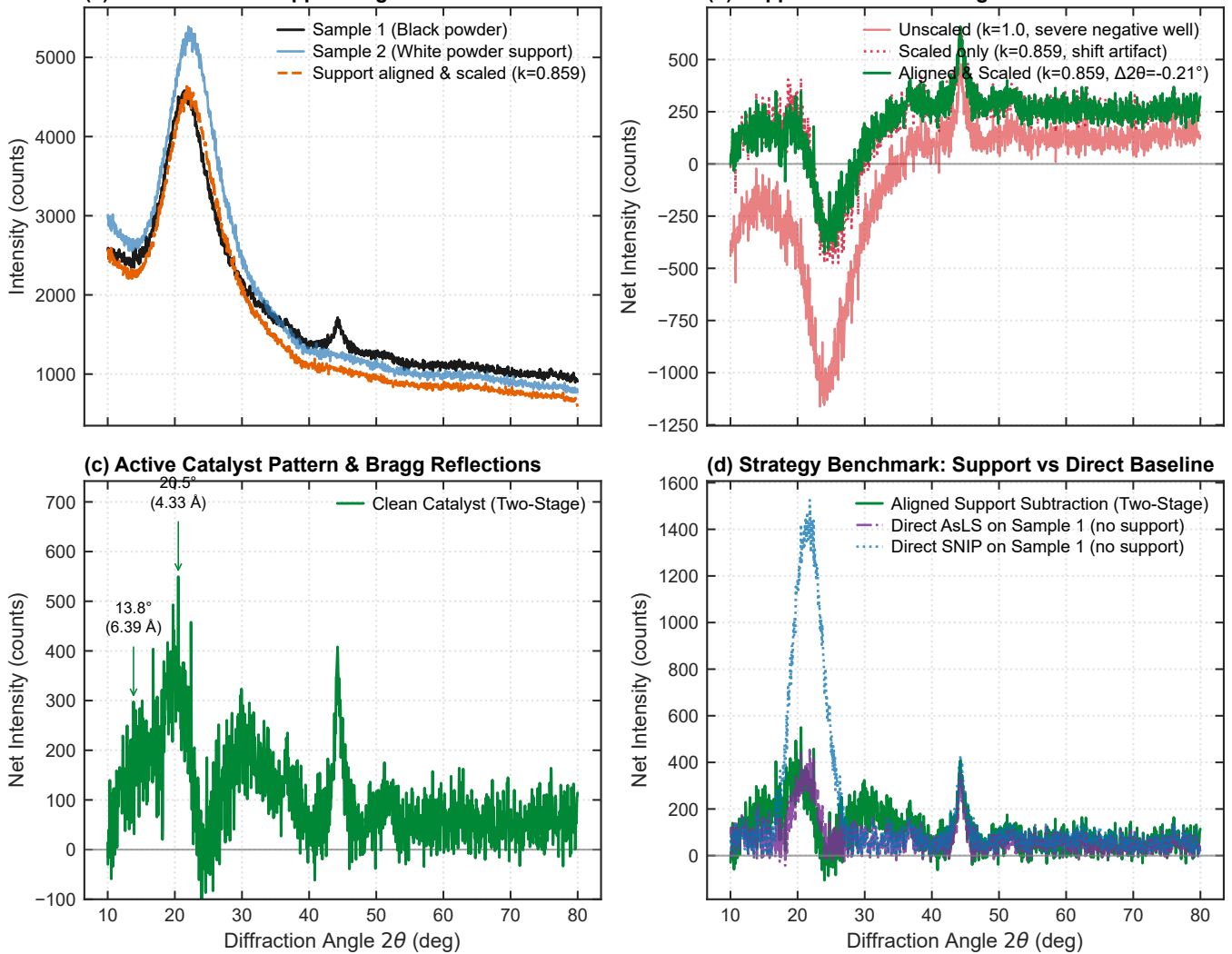

(a) Raw Profiles & Support Alignment
Sample 1 (Black powder)
Sample 2 (White powder support)
Support aligned & scaled (k=0.859)
Intensity (counts)
(b) Support Subtraction Diagnostics
Unscaled (k=1.0, severe negative well)
Scaled only (k=0.859, shift artifact)
Aligned & Scaled (k=0.859, Δ2θ=-0.21°)
Net Intensity (counts)
(c) Active Catalyst Pattern & Bragg Reflections
20.5°
(4.33 Å)
13.8°
(6.39 Å)
Clean Catalyst (Two-Stage)
Net Intensity (counts)
Diffraction Angle 2θ (deg)
(d) Strategy Benchmark: Support vs Direct Baseline
Aligned Support Subtraction (Two-Stage)
Direct AsLS on Sample 1 (no support)
Direct SNIP on Sample 1 (no support)
Net Intensity (counts)
Diffraction Angle 2θ (deg)